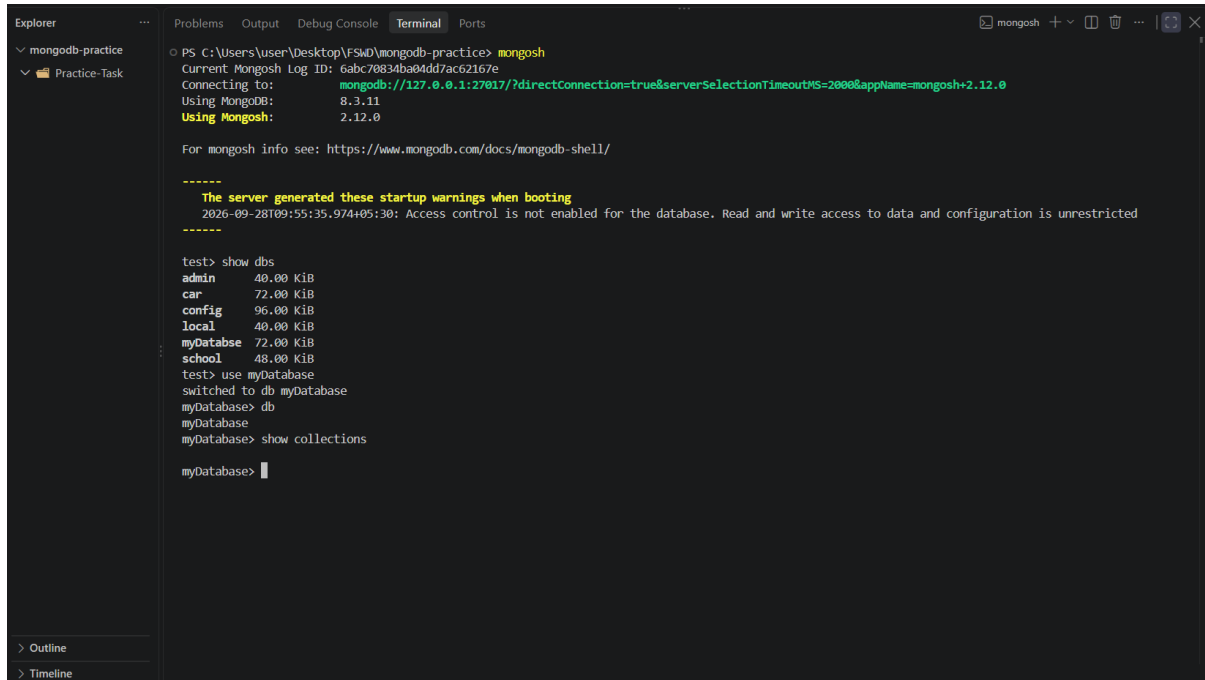


MongoDB Practice

1. Basic Commands:

Mongosh, show dbs, use <dbname>, db



```
PS C:\Users\User\Desktop\FSD\mongodb-practice> mongosh
Current Mongosh Log ID: 6abc70834ba04dd7ac62167e
Connecting to: mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000&appName=mongosh+2.12.0
Using MongoDB: 8.3.11
Using Mongosh: 2.12.0

For mongosh info see: https://www.mongodb.com/docs/mongosh-shell/

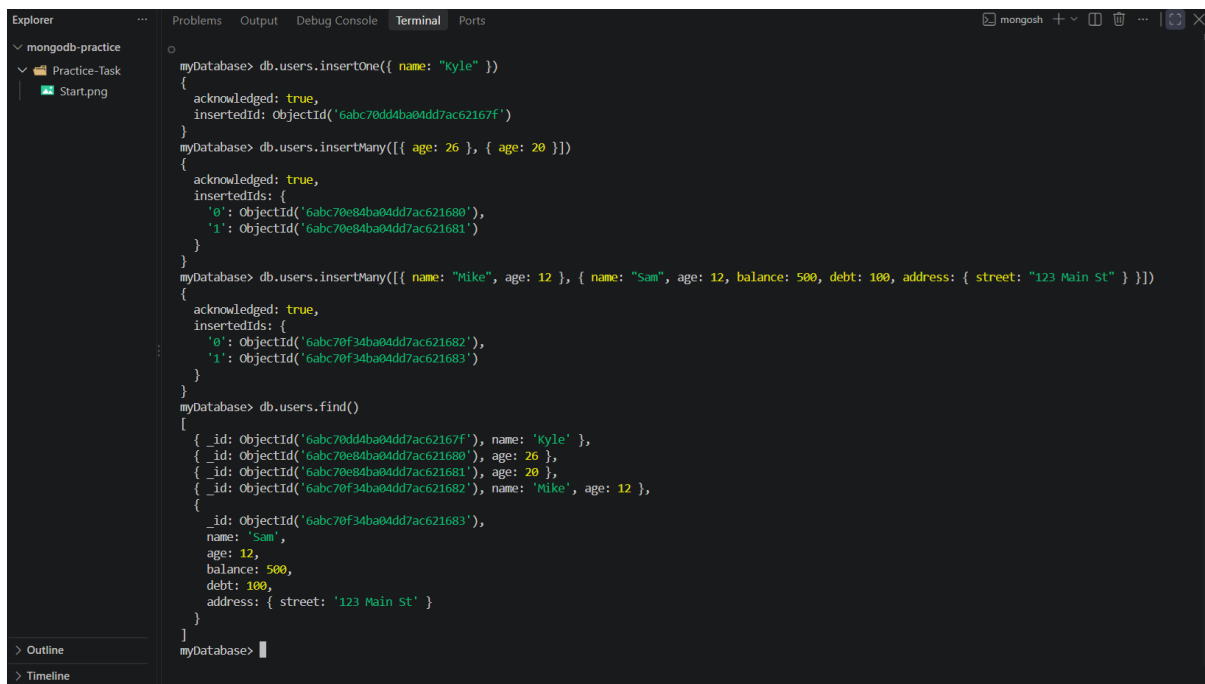
-----
The server generated these startup warnings when booting
2026-09-28T09:55:35.974+05:30: Access control is not enabled for the database. Read and write access to data and configuration is unrestricted
-----

test> show dbs
admin          40.00 KiB
car            72.00 KiB
config         96.00 KiB
local          40.00 KiB
myDatabase     72.00 KiB
school         48.00 KiB
test> use myDatabase
switched to db myDatabase
myDatabase> db
myDatabase
myDatabase> show collections

myDatabase>
```

2. Create:

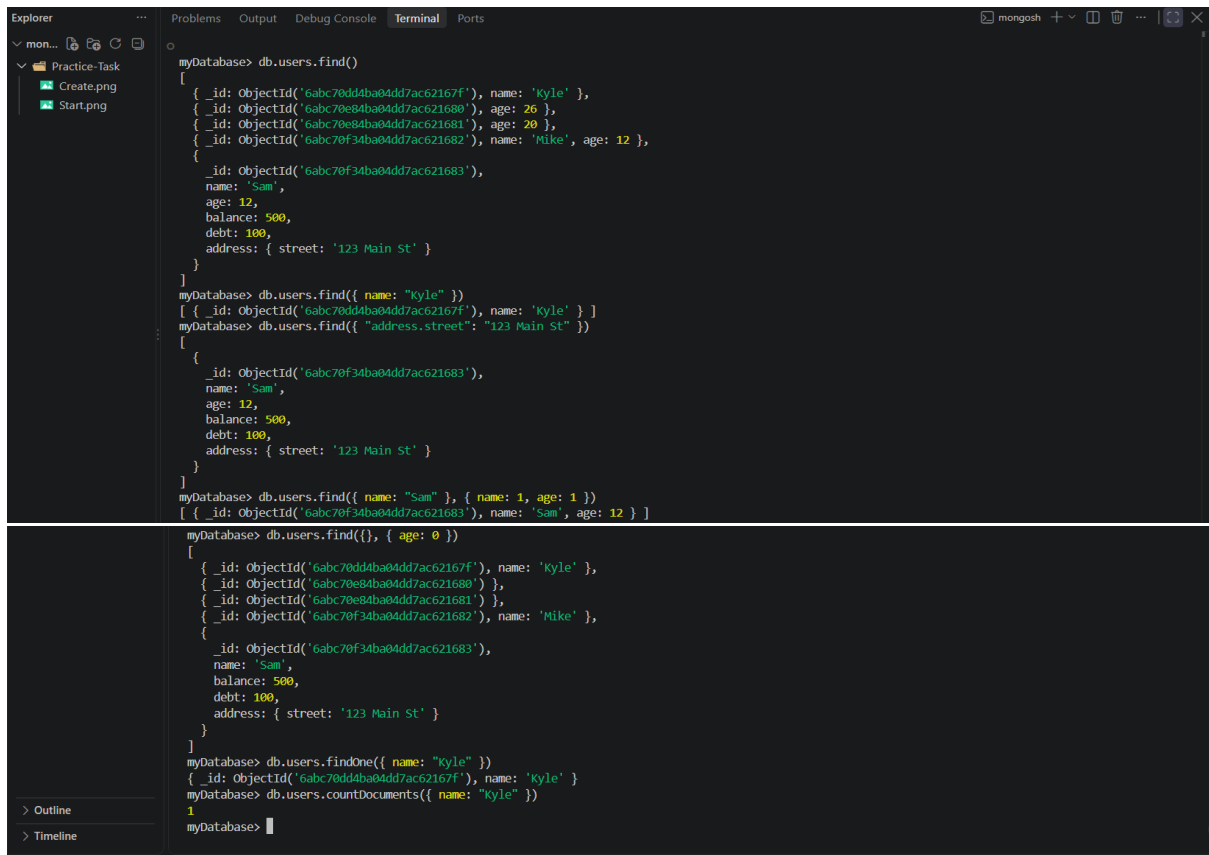
insertOne, insertMany



```
myDatabase> db.users.insertOne({ name: "Kyle" })
{ acknowledged: true,
  insertedId: ObjectId('6abc70dd4ba04dd7ac62167f') }
myDatabase> db.users.insertMany([ { age: 26 }, { age: 20 } ])
{ acknowledged: true,
  insertedIds: {
    '0': ObjectId('6abc70e84ba04dd7ac621680'),
    '1': ObjectId('6abc70e84ba04dd7ac621681')
  } }
myDatabase> db.users.insertMany([ { name: "Mike", age: 12 }, { name: "Sam", age: 12, balance: 500, debt: 100, address: { street: "123 Main St" } } ])
{ acknowledged: true,
  insertedIds: {
    '0': ObjectId('6abc70f34ba04dd7ac621682'),
    '1': ObjectId('6abc70f34ba04dd7ac621683')
  } }
myDatabase> db.users.find()
[
  { _id: ObjectId('6abc70dd4ba04dd7ac62167f'), name: 'Kyle' },
  { _id: ObjectId('6abc70e84ba04dd7ac621680'), age: 26 },
  { _id: ObjectId('6abc70e84ba04dd7ac621681'), age: 20 },
  { _id: ObjectId('6abc70f34ba04dd7ac621682'), name: 'Mike', age: 12 },
  {
    _id: ObjectId('6abc70f34ba04dd7ac621683'),
    name: 'Sam',
    age: 12,
    balance: 500,
    debt: 100,
    address: { street: '123 Main St' }
  }
]
myDatabase>
```

3. Read:

Find, find(<filterObject>), find(<filterObject>, <selectObject>), findOne, countDocuments



```
myDatabase> db.users.find()
[ { _id: ObjectId('6abc70dd4ba04dd7ac62167f'), name: 'Kyle' },
  { _id: ObjectId('6abc70e84ba04dd7ac621680'), age: 26 },
  { _id: ObjectId('6abc70e84ba04dd7ac621681'), age: 20 },
  { _id: ObjectId('6abc70f34ba04dd7ac621682'), name: 'Mike', age: 12 },
  {
    _id: ObjectId('6abc70f34ba04dd7ac621683'),
    name: 'Sam',
    age: 12,
    balance: 500,
    debt: 100,
    address: { street: '123 Main St' }
  }
]

myDatabase> db.users.find({ name: "Kyle" })
[ { _id: ObjectId('6abc70dd4ba04dd7ac62167f'), name: 'Kyle' } ]

myDatabase> db.users.find({ "address.street": "123 Main St" })
[ {
  _id: ObjectId('6abc70f34ba04dd7ac621683'),
  name: 'Sam',
  age: 12,
  balance: 500,
  debt: 100,
  address: { street: '123 Main St' }
} ]

myDatabase> db.users.find({ name: "Sam" }, { name: 1, age: 1 })
[ { _id: ObjectId('6abc70f34ba04dd7ac621683'), name: 'Sam', age: 12 } ]

myDatabase> db.users.find({}, { age: 0 })
[ { _id: ObjectId('6abc70dd4ba04dd7ac62167f'), name: 'Kyle' },
  { _id: ObjectId('6abc70e84ba04dd7ac621680'), age: 26 },
  { _id: ObjectId('6abc70e84ba04dd7ac621681'), age: 20 },
  { _id: ObjectId('6abc70f34ba04dd7ac621682'), name: 'Mike', age: 12 },
  {
    _id: ObjectId('6abc70f34ba04dd7ac621683'),
    name: 'Sam',
    balance: 500,
    debt: 100,
    address: { street: '123 Main St' }
  }
]

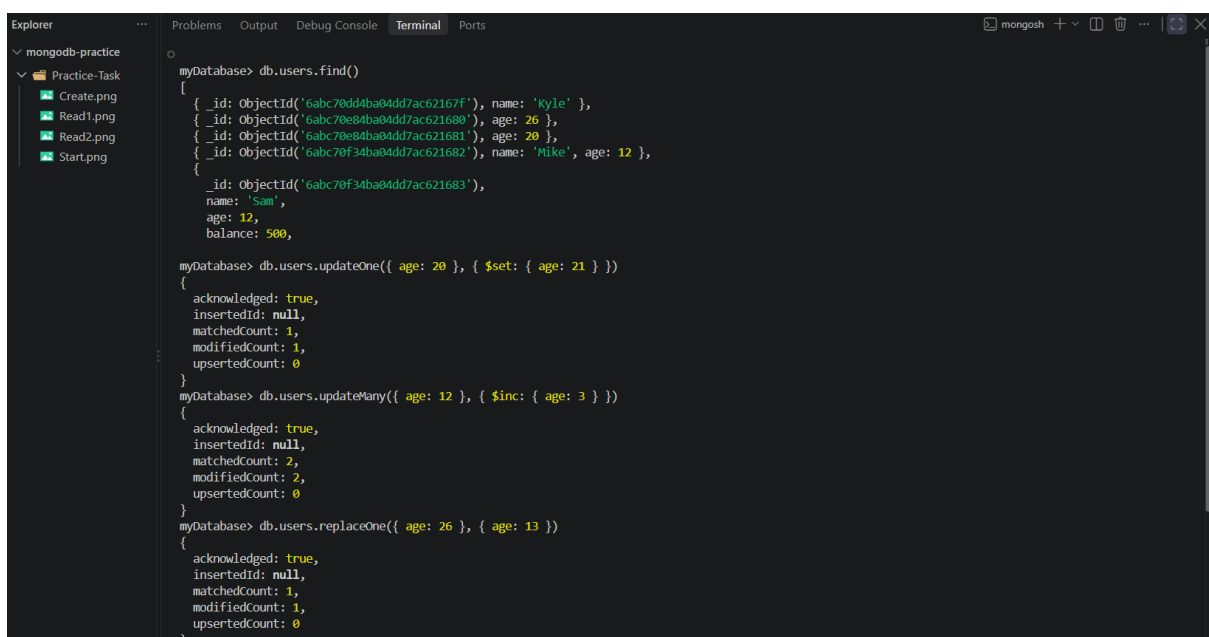
myDatabase> db.users.findOne({ name: "Kyle" })
{ _id: ObjectId('6abc70dd4ba04dd7ac62167f'), name: 'Kyle' }

myDatabase> db.users.countDocuments({ name: "Kyle" })
1

myDatabase>
```

4. Update:

updateOne, updateMany, replaceOne



```
myDatabase> db.users.find()
[ { _id: ObjectId('6abc70dd4ba04dd7ac62167f'), name: 'Kyle' },
  { _id: ObjectId('6abc70e84ba04dd7ac621680'), age: 26 },
  { _id: ObjectId('6abc70e84ba04dd7ac621681'), age: 20 },
  { _id: ObjectId('6abc70f34ba04dd7ac621682'), name: 'Mike', age: 12 },
  {
    _id: ObjectId('6abc70f34ba04dd7ac621683'),
    name: 'Sam',
    age: 12,
    balance: 500,
  }
]

myDatabase> db.users.updateOne({ age: 20 }, { $set: { age: 21 } })
{ acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0 }

myDatabase> db.users.updateMany({ age: 12 }, { $inc: { age: 3 } })
{ acknowledged: true,
  insertedId: null,
  matchedCount: 2,
  modifiedCount: 2,
  upsertedCount: 0 }

myDatabase> db.users.replaceOne({ age: 26 }, { age: 13 })
{ acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0 }

myDatabase>
```

```
myDatabase> db.users.find()
[
  { _id: ObjectId('6abc70dd4ba04dd7ac62167f'), name: 'Kyle' },
  { _id: ObjectId('6abc70e84ba04dd7ac621680'), age: 13 },
  { _id: ObjectId('6abc70e84ba04dd7ac621681'), age: 21 },
  { _id: ObjectId('6abc70f34ba04dd7ac621682'), name: 'Mike', age: 15 },
  {
    _id: ObjectId('6abc70f34ba04dd7ac621683'),
    name: 'Sam',
    age: 15,
    balance: 500,
    debt: 100,
    address: { street: '123 Main St' }
  }
]
myDatabase> 
```

5. Delete:

deleteOne, deleteMany

```
myDatabase> db.users.updateOne({ age: 20 }, { $set: { age: 21 } })
{ acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0 }
myDatabase> db.users.updateMany({ age: 12 }, { $inc: { age: 3 } })
{ acknowledged: true,
  insertedId: null,
  matchedCount: 2,
  modifiedCount: 2,
  upsertedCount: 0 }
myDatabase> db.users.replaceOne({ age: 26 }, { age: 13 })
{ acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0 }
myDatabase> db.users.find()
[
  { _id: ObjectId('6abc76e28f44276825a16af0'), name: 'Kyle' },
  { _id: ObjectId('6abc76e28f44276825a16af1'), age: 13 },
  { _id: ObjectId('6abc76e28f44276825a16af2'), age: 21 },
  {
    _id: ObjectId('6abc76e28f44276825a16af4'),
    name: 'Sam',
    age: 15,
    balance: 500,
    debt: 100,
    address: { street: '123 Main St' }
  }
]
myDatabase> db.users.deleteOne({ age: 15 })
{ acknowledged: true, deletedCount: 1 }
myDatabase> db.users.find()
[
  { _id: ObjectId('6abc76e28f44276825a16af0'), name: 'Kyle' },
  { _id: ObjectId('6abc76e28f44276825a16af1'), age: 13 },
  { _id: ObjectId('6abc76e28f44276825a16af2'), age: 21 },
  {
    _id: ObjectId('6abc76e28f44276825a16af4'),
    name: 'Sam',
    age: 15,
    balance: 500,
    debt: 100,
    address: { street: '123 Main St' }
  }
]
myDatabase> db.users.deleteMany({ age: 15 })
{ acknowledged: true, deletedCount: 1 }
myDatabase> db.users.find()
[
  { _id: ObjectId('6abc76e28f44276825a16af0'), name: 'Kyle' },
  { _id: ObjectId('6abc76e28f44276825a16af1'), age: 13 },
  { _id: ObjectId('6abc76e28f44276825a16af2'), age: 21 }
]
myDatabase> 
```

6. Complex Filter Object:

\$eq, \$ne, \$gt / \$gte, \$lt / \$lte, \$in, \$nin, \$and, \$or, \$not, \$exists, \$expr



```
myDatabase> db.users.updateOne({ age: 20 }, { $set: { age: 21 } })
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
myDatabase> db.users.updateMany({ age: 12 }, { $inc: { age: 3 } })
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 2,
  modifiedCount: 2,
  upsertedCount: 0
}
myDatabase> db.users.replaceOne({ age: 26 }, { age: 13 })
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
myDatabase> db.users.find()
[
  { _id: ObjectId('6abc76e28f44276825a16af0'), name: 'Kyle' },
]
myDatabase> db.users.find({ name: { $eq: "Kyle" } })
[ { _id: ObjectId('6abc76e28f44276825a16af0'), name: 'Kyle' } ]
myDatabase> db.users.find({ name: { $ne: "Kyle" } })
[
  { _id: ObjectId('6abc76e28f44276825a16af1'), age: 13 },
  { _id: ObjectId('6abc76e28f44276825a16af2'), age: 21 }
]

myDatabase> db.users.find({ age: { $gt: 12 } })
[
  { _id: ObjectId('6abc76e28f44276825a16af1'), age: 13 },
  { _id: ObjectId('6abc76e28f44276825a16af2'), age: 21 }
]
myDatabase> db.users.find({ age: { $gte: 15 } })
[ { _id: ObjectId('6abc76e28f44276825a16af2'), age: 21 } ]
myDatabase> db.users.find({ age: { $lt: 20 } })
[ { _id: ObjectId('6abc76e28f44276825a16af1'), age: 13 } ]
myDatabase> db.users.find({ age: { $lte: 15 } })
[ { _id: ObjectId('6abc76e28f44276825a16af1'), age: 13 } ]
myDatabase> db.users.find({ name: { $in: ["Kyle", "Mike"] } })
[ { _id: ObjectId('6abc76e28f44276825a16af0'), name: 'Kyle' } ]
myDatabase> db.users.find({ name: { $nin: ["Kyle", "Mike"] } })
[
  { _id: ObjectId('6abc76e28f44276825a16af1'), age: 13 },
  { _id: ObjectId('6abc76e28f44276825a16af2'), age: 21 }
]
myDatabase> db.users.find({ $and: [{ age: 15 }, { name: "Sam" }] })

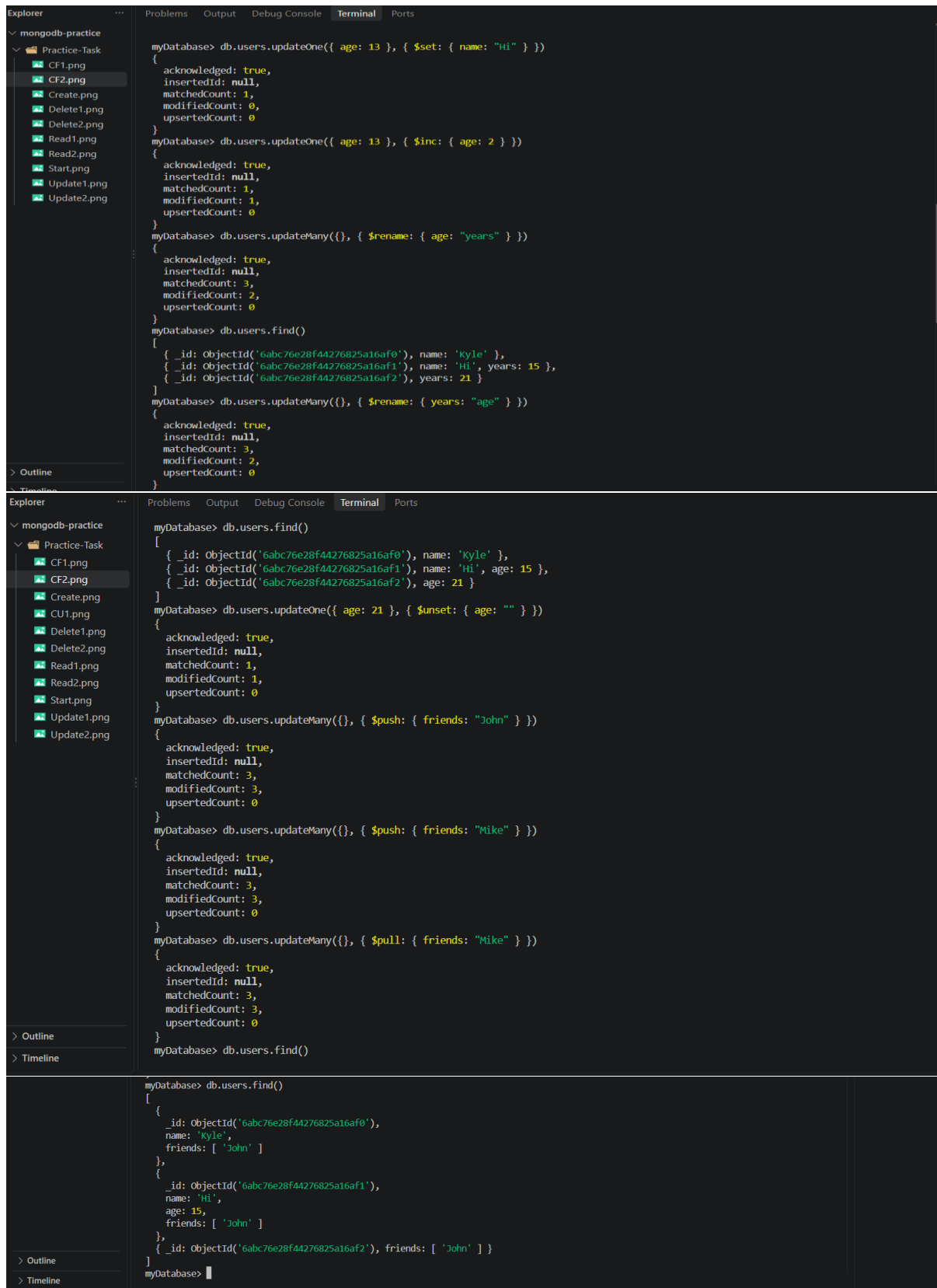
myDatabase> db.users.find({ age: 15, name: "Sam" })

myDatabase> db.users.find({ $or: [{ age: 15 }, { name: "kyle" }] })
[ { _id: ObjectId('6abc76e28f44276825a16af0'), name: 'Kyle' } ]
myDatabase> db.users.find({ name: { $not: { $eq: "Kyle" } } })
[
  { _id: ObjectId('6abc76e28f44276825a16af1'), age: 13 },
  { _id: ObjectId('6abc76e28f44276825a16af2'), age: 21 }
]
myDatabase> db.users.find({ name: { $exists: true } })
[ { _id: ObjectId('6abc76e28f44276825a16af0'), name: 'Kyle' } ]
myDatabase> db.users.find({ $expr: { $gt: [{"$balance", "$debt"}] } })

myDatabase> []
```

7. Complex Update Object:

\$set, \$inc, \$rename, \$unset, \$push, \$pull



The screenshot displays a MongoDB terminal session with three panels. The left panel shows the Explorer view with a file tree for 'mongodb-practice'. The middle panel shows the Terminal view with the following commands and output:

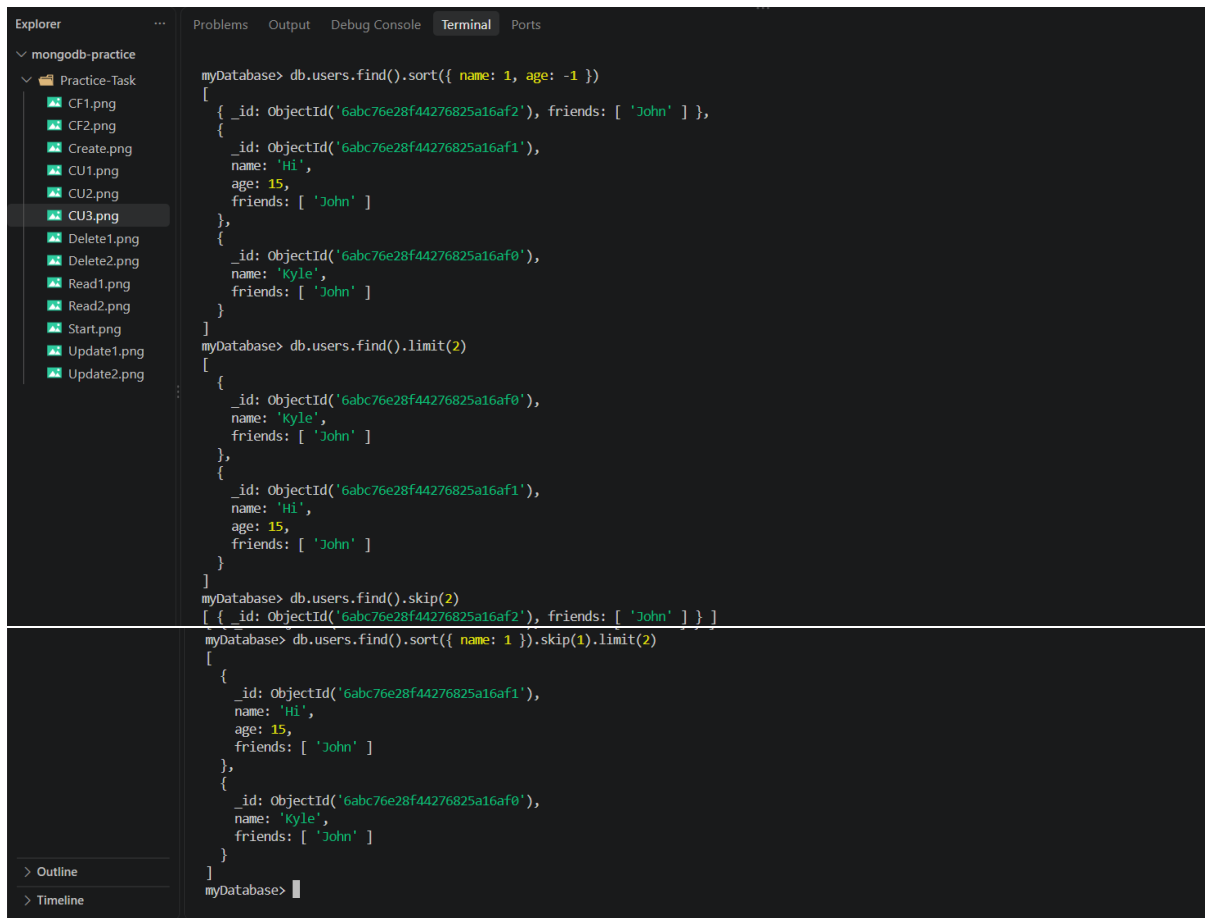
```
myDatabase> db.users.updateOne({ age: 13 }, { $set: { name: "Hi" } })
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 0,
  upsertedCount: 0
}
myDatabase> db.users.updateOne({ age: 13 }, { $inc: { age: 2 } })
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
myDatabase> db.users.updateMany({}, { $rename: { age: "years" } })
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 3,
  modifiedCount: 2,
  upsertedCount: 0
}
myDatabase> db.users.find()
[
  { _id: ObjectId('6abc76e28f44276825a16af0'), name: 'Kyle' },
  { _id: ObjectId('6abc76e28f44276825a16af1'), name: 'Hi', years: 15 },
  { _id: ObjectId('6abc76e28f44276825a16af2'), years: 21 }
]
myDatabase> db.users.updateMany({}, { $rename: { years: "age" } })
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 3,
  modifiedCount: 2,
  upsertedCount: 0
}
```

The right panel shows the same terminal session with the following commands and output:

```
myDatabase> db.users.find()
[
  { _id: ObjectId('6abc76e28f44276825a16af0'), name: 'Kyle' },
  { _id: ObjectId('6abc76e28f44276825a16af1'), name: 'Hi', age: 15 },
  { _id: ObjectId('6abc76e28f44276825a16af2'), age: 21 }
]
myDatabase> db.users.updateOne({ age: 21 }, { $unset: { age: "" } })
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
myDatabase> db.users.updateMany({}, { $push: { friends: "John" } })
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 3,
  modifiedCount: 3,
  upsertedCount: 0
}
myDatabase> db.users.updateMany({}, { $push: { friends: "Mike" } })
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 3,
  modifiedCount: 3,
  upsertedCount: 0
}
myDatabase> db.users.updateMany({}, { $pull: { friends: "Mike" } })
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 3,
  modifiedCount: 3,
  upsertedCount: 0
}
myDatabase> db.users.find()
[
  {
    _id: ObjectId('6abc76e28f44276825a16af0'),
    name: 'Kyle',
    friends: [ 'John' ]
  },
  {
    _id: ObjectId('6abc76e28f44276825a16af1'),
    name: 'Hi',
    age: 15,
    friends: [ 'John' ]
  },
  {
    _id: ObjectId('6abc76e28f44276825a16af2'), friends: [ 'John' ]
  }
]
```

8. Read Modifiers:

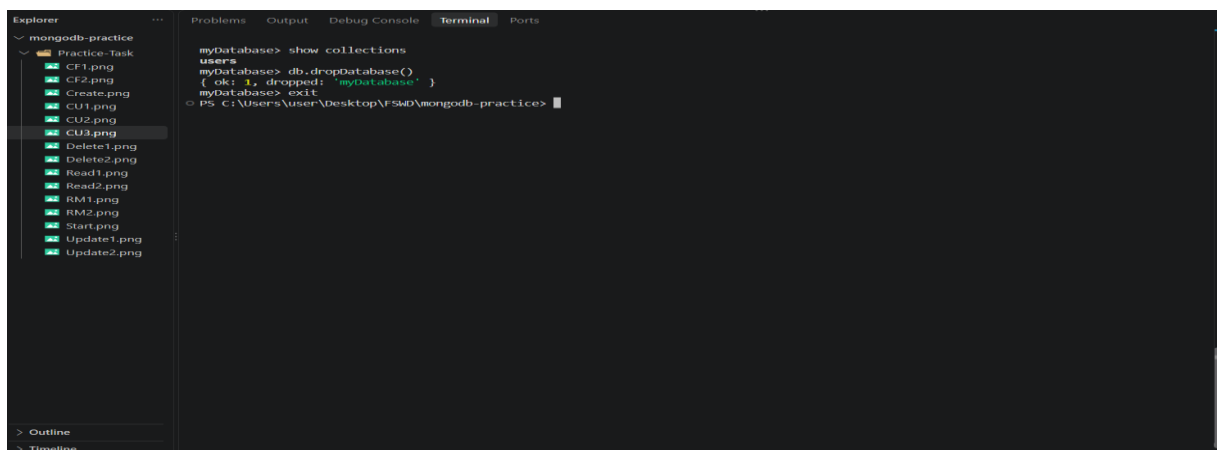
Sort, limit, skip



```
myDatabase> db.users.find().sort({ name: 1, age: -1 })
[
  { _id: ObjectId('6abc76e28f44276825a16af2'), friends: [ 'John' ] },
  {
    _id: ObjectId('6abc76e28f44276825a16af1'),
    name: 'Hi',
    age: 15,
    friends: [ 'John' ]
  },
  {
    _id: ObjectId('6abc76e28f44276825a16af0'),
    name: 'Kyle',
    friends: [ 'John' ]
  }
]
myDatabase> db.users.find().limit(2)
[
  {
    _id: ObjectId('6abc76e28f44276825a16af0'),
    name: 'Kyle',
    friends: [ 'John' ]
  },
  {
    _id: ObjectId('6abc76e28f44276825a16af1'),
    name: 'Hi',
    age: 15,
    friends: [ 'John' ]
  }
]
myDatabase> db.users.find().skip(2)
[ { _id: ObjectId('6abc76e28f44276825a16af2'), friends: [ 'John' ] } ]
myDatabase> db.users.find().sort({ name: 1 }).skip(1).limit(2)
[
  {
    _id: ObjectId('6abc76e28f44276825a16af1'),
    name: 'Hi',
    age: 15,
    friends: [ 'John' ]
  },
  {
    _id: ObjectId('6abc76e28f44276825a16af0'),
    name: 'Kyle',
    friends: [ 'John' ]
  }
]
myDatabase>
```

9. Basic Commands:

Show collections, db.dropDatabase(), exit



```
myDatabase> show collections
users
myDatabase> db.dropDatabase()
{ ok: 1, dropped: 'myDatabase' }
myDatabase> exit
PS C:\Users\User\Desktop\FSD>mongodb-practice>
```